Week 10 Notes & Quiz

Notes for all seven Week 10 lectures, made only from the 221 slides in *WEEK 10.pdf*. Each yellow tag such as S62 is the slide (PDF page) the point comes from. The 59-question practice questions and answer key are at the end.

STOP	LECTURE	SLIDES
L45	**Demand Forecasting**	S1–S45
L46	**Capacity Planning**	S46–S68
L47	**Sequencing & Scheduling**	S69–S97
L48	**Transportation Management**	S98–S128
L49	**Transportation Problem Programming: NWCR**	S129–S151
L50	**Transportation Problem Programming: LCM, VAM, MODI**	S152–S193
L51	**Assignment Problem Programming**	S194–S221
+	**Cheat sheet · Slide quirks · Quiz (59 Qs)**	—

LECTURE 45 S1–S45

Demand Forecasting

Indian demand context

- World Bank India Development Update (Oct 2023): India grew **7.2%** in FY22/23, **second highest in the G20**, almost twice the emerging-market average; bank credit growth 15.8%; FY23/24 forecast **6.3%** (services 7.4%, investment 8.9%). S3
- Q3 FY2024: real GDP **8.4%**; private investment +10.6%; private consumption 3.5%; government consumption −3.2%. S4
- GVA grew **6.5%** (manufacturing 11.6%, construction 9.5%, services 7%, agriculture −0.8%). GDP growing faster than GVA means **demand outpacing supply**, a possible source of inflation. S5

> DEMAND FORECASTING S7
>
> Understanding and predicting customer demand to make decisions on supply chain operations, margins, cash flow, capex and capacity planning, often using historical data; it helps avoid over- and under-production.

- Five influencing factors: **seasonality, competition, geography, economy, types of goods** (perishables need precise forecasts; subscriptions are predictable). S8–S9
- Advantages: inventory management, cost reduction, customer satisfaction, production planning, resource allocation, marketing, financial planning, risk mitigation, strategic decisions, SC optimization, seasonal planning, competitive advantage. S10–S12
- Challenges: lack of data, over-reliance on past data, limited visibility, inaccurate models, poor processes, insufficient resources, unpredictable external factors. S13–S14

TYPE	MEANING S15–S16
Passive	Uses only past sales; for firms aiming at **steady** sales.
Active	For startups and fast-growing firms; includes growth plans and competition.
Short-term	Day-to-day, e.g. Black Friday inventory; JIT chains.
Long-term	**More than a year**; seasonality, capacity, expansion.
Macro / micro	Macro: economy, competition, consumer trends. Micro: a specific segment (organic peanut butter for millennial parents in Austin).

Techniques

Qualitative: Delphi, salesforce composite, expert/insider knowledge, surveys, interviews/focus groups. **Quantitative**: moving averages, trend projection, seasonal averages, econometric models, machine learning. S17

- **Delphi**: panel of experts (typically **10–50**) answer questionnaires independently (avoids groupthink); summaries fed back; repeated for **2–4 rounds** until consensus. S18–S19
- **Salesforce composite**: combine CRM data, several methods with weights, and sales reps' input. S20–S21
- Surveys: online, telephone, face-to-face (high response, costly), mail (low response). S24–S26

Moving averages (sales Jan–Jun: 150, 160, 170, 180, 190, 200)

METHOD	RULE	SLIDE RESULT
SMA (3-month)	Average of last 3 periods	`Mar 160 · Apr 170 · May 180 · Jun 190` S30
EMA	$EMA_t = \alpha \cdot Y_t + (1-\alpha) \cdot EMA_{t-1}$; $\alpha = 2/(N+1) =$ **0.5** for N = 3; start $EMA_{Jan} = 150$	`Feb 155 · Mar 162.5 · Apr 171.25 · May 180.625 · Jun 190.3125` S32–S35
WMA	Weights 1, 2, 3, 4 (latest highest)	`(150 + 320 + 510 + 720) ÷ 10 = 170` S37

- EMA gives more weight to recent data and reacts faster; WMA filters outliers; shorter SMA is more reactive. S29–S36
- **Trend projection**: collect data → plot → choose model → fit → extrapolate → evaluate. Models: linear $Y = a + bt$, exponential $Y = ae^{bt}$, polynomial. S38–S41
- Regression: $b_1 = \Sigma(x - \bar{x})(y - \bar{y}) \div \Sigma(x - \bar{x})^2$; $b_0 = \bar{y} - b_1\bar{x}$. TV ads (1, 3, 2, 1, 3) vs cars sold (14, 24, 18, 17, 27) → $y = 5x + 10$. S42

LECTURE 46 S46–S68

Capacity Planning

- Capacity planning balances available resources with demand; capacity = work completed in a given time. S48
- By horizon: **long-term/strategic** (several years), **medium-term/tactical** (months to a year), **short-term/operational** (days to months, reactive). S49
- By resource: workforce, product, tool, resource, project capacity planning. S53–S54

Design capacity	Maximum output under ideal conditions. S50
Effective (system) capacity	Design capacity minus allowances (maintenance, breaks, scheduling). S50–S51
Actual output	Cannot exceed effective capacity; hit by breakdowns, absenteeism, shortages, demand. S50
Efficiency	**Actual output ÷ effective capacity × 100** S51
Utilization	**Actual output ÷ design capacity × 100** S51

EXAMPLE 1 S52

230 units/hour, 7 days × 3 shifts × 8 h → design capacity = 168 × 230 = **38,640**/week. Effective 28,000; actual 25,000. Utilization = 25,000 ÷ 38,640 = **64.7%**; efficiency = 25,000 ÷ 28,000 = **89.3%**.

- Process: estimate requirements → evaluate existing capacity and gaps → identify alternatives → financial analysis → qualitative issues → choose the best long-term option → implement → monitor. S55–S57
- Alternatives: add capacity, optimize existing, outsource, flexible workforce, demand management, contingency plans, Industry 4.0. S56
- Demand patterns: growth, decline, cyclical, stable. S58
- Evaluating alternatives: **economic** (break-even analysis) and **non-economic** (public opinion, employee and community reaction). S59

Break-even analysis

TC = FC + VC; VC = Q × v; TR = Q × r; P = Q(r − v) − FC; $Q_{BEP} = FC \div (r - v)$. S60–S61

EXAMPLE	WORKING
Pizza: FC $4,000/month, v $4, r $9 S62	BEP = 4,000 ÷ 5 = **800** pizzas. Profit $10,000 needs (10,000 + 4,000) ÷ 5 = **2,800**. Demand of 700 < 800 → **not profitable**.
Make or buy (ABX): make FC $160,000, v $100; buy $150/unit S63–S65	$Q = FC \div (V_B - V_M) = 160{,}000 \div 50 =$ **3,200**. At 1,000 units: make $260,000 vs buy $150,000 → **buy**. Make is cheaper for Q > 3,200.

LECTURE 47 S69–S97

Sequencing & Scheduling

- **Sequencing**: the order of processing tasks on resources. **Scheduling**: sequencing plus start and finish times. S71

Sequencing rules on one example

Jobs A–E: processing 3, 4, 2, 6, 1 days; due 5, 6, 7, 9, 2. S73

RULE	ORDER	TOTAL FLOW	MEAN FLOW	AVERAGE DELAY
FIFO	A B C D E	50	10	4.6 S73
LIFO	E D C B A	46	9.2	4.0 S74
SPT	E C A B D	36	7.2	2.4 S75
LPT	D B A C E	60	12	6.8 S76
EDD	E A B C D	39	7.8	2.4 S77

- **SPT** gives the lowest total and mean flow time; **LPT** aims to cut changeovers and raise machine utilization; **EDD** targets deadlines. S75–S77
- **Least slack (LS)**: slack s = d − p. Jobs (p, d): J1 (4, 10), J2 (3, 8), J3 (5, 12), J4 (2, 7), J5 (1, 5) → slacks 6, 5, 7, 5, 4 → sequence **J5, J2, J4, J1, J3**. S78–S79
- **Critical ratio** = time remaining until due date ÷ remaining processing time; lowest first; combines EDD and SPT; dynamic. A 10/5 = 2.0, B 15/3 = 5.0, C 5/2 = 2.5, D 7/7 = 1.0 → **D, A, C, B**. S80–S82
- Advantages: avoid bottlenecks, adaptable, lean, archive know-how. Challenges: dynamic changes, limited resources, complexity, conflicting priorities, setup times, technology limits. S83–S84

Scheduling and Johnson's rule

- Single-machine scheduling: n jobs, one operation. **Flow shop**: n jobs, m operations, same sequence. General flow shop is exponential; Johnson's rule solves **2 machines, n jobs** optimally in polynomial time. S85–S86
- Rule: pick the shortest time; if on machine 1 schedule the job **first**, if on machine 2 schedule it **last**; remove it; repeat. Conditions: constant times, independent of sequence, equal priority, all jobs go M1 then M2. S87
- Example (M1, M2): A (3.2, 4.2), B (4.7, 1.5), C (2.2, 5.0), D (5.8, 4.0), E (3.1, 2.8) → C → A → D → E → B. S88–S90
- **Three machines**: if min(A) ≥ max(B) or min(C) ≥ max(B), replace with G = A + B and H = B + C and apply Johnson's rule. S91

Item	Machine A		Machine B		Machine C	
	Time In	Time Out	Time In	Time Out	Time In	Time Out
1	0	4	4	8	8	14
4	4	10	10	12	14	22
5	10	13	13	19	22	29
3	13	21	21	24	29	40
2	21	30	30	35	40	49

Total elapsed time = 49

Idle time for machine A = 49 − 30 = 19 hours

Idle time for machine B = 4 + (10 - 8) + (13 - 12) + (21 - 19)+ (30 - 24) + (49 - 35) = 29 hours

Idle time for machine C = 8 + (14 - 14) + (22 - 22) + (29 - 29)+ (40 - 40) = 8 hours

NPTEL

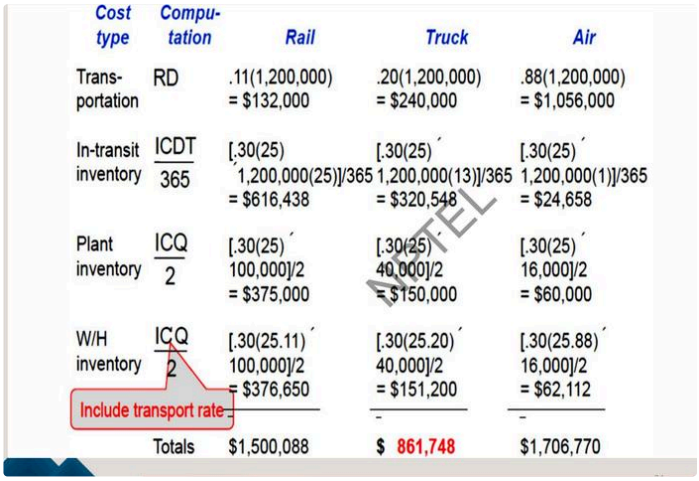
Original slide S94, MDH Masala (3 machines): min A = 3, max B = 6, min C = 6 → condition holds. Sequence **1, 4, 5, 3, 2**; total elapsed time **49**; idle time A 19, B 29, C 8 hours.

LECTURE 48 S98–S128

Transportation Management

- Four major components of transportation decisions: **flow, mode, load, route**. S100
- **Load**: quantity and composition of goods per trip. Aspects: size, weight (regulations, fines), volume (palletization, stacking), packaging, composition (consolidation), **balancing** (axle weight, centre of gravity). S101–S104
- Fleet size: transport demand, service levels (SLAs), geography, cargo characteristics, redundancy, total cost of ownership, technology (telematics), regulation, scalability. S105–S107
- Mode choice factors: transit time, cost, reliability, distance, safety, environment. S110

MODE	BEST FOR · DRAWBACK S111–S116
Road	Door-to-door, short-medium distance, last mile, JIT · congestion, higher cost per mile long-haul
Rail	Long distance, bulk, heavy, energy-efficient · limited access, extra handling
Maritime	Bulk, long-haul, **lowest cost per ton**, global trade · slow, port congestion
Air	Time-sensitive, high-value, perishable · highest cost, limited bulk
Pipeline	Liquids, gases; low energy · high initial investment, limited cargo types
Intermodal	Combines modes; door-to-door · coordination and documentation



Cost type	Computation	Rail	Truck	Air
Transportation	RD	.11(1,200,000) = $132,000	.20(1,200,000) = $240,000	.88(1,200,000) = $1,056,000
In-transit inventory	ICDT / 365	[.30(25) 1,200,000(25)]/365 = $616,438	[.30(25) 1,200,000(13)]/365 = $320,548	[.30(25) 1,200,000(1)]/365 = $24,658
Plant inventory	ICQ / 2	[.30(25) 100,000]/2 = $375,000	[.30(25) 40,000]/2 = $150,000	[.30(25) 16,000]/2 = $60,000
W/H inventory	ICQ / 2	[.30(25.11) 100,000]/2 = $376,650	[.30(25.20) 40,000]/2 = $151,200	[.30(25.88) 16,000]/2 = $62,112
	Totals	$1,500,088	$ 861,748	$1,706,770

Include transport rate

Original slide S118. Ship 1,200,000 lb/yr worth $25/lb, carrying cost 30%. Rail $0.11/lb, 25 days, 100,000 lb lots; truck $0.20, 13 days, 40,000 lb; air $0.88, 1 day, 16,000 lb. Total = transport + in-transit + plant + warehouse inventory: rail **$1,500,088**, **truck $861,748 (lowest)**, air $1,706,770.

- Logistics flows: physical, information, financial, reverse. **Route planning** weighs distance, time, cost, capacity, mode, infrastructure, regulation. S119–S123
- US firms spend over **$600 billion** a year on transportation and about **$1 billion** on **TMS** (software that automates carrier selection, load building, routing, scheduling, freight audit). S125
- TMS examples: Polaroid (HighJump), **Tree of Life** inbound freight efficiency **55% → 75%** (JDA), Orange Glo (LeanLogistics). ARC Advisory Group: TMS value now spans other departments and partners. S126

LECTURE 49 S129–S151

Transportation Problem: Basics and North-West Corner

- A special **linear programming** problem: ship one commodity from **m sources** to **n destinations** at minimum total cost, meeting supply and demand. S131 S134
- Assumptions: fixed supply and demand, single commodity, **linear costs**, route capacity, non-negative variables, full shipment. S132–S133
- **Balanced**: total supply = total demand. **Unbalanced**: add a dummy source or destination. S138
- Phase I: initial basic feasible solution (NWCR, LCM, VAM). Phase II: optimal solution (MODI). A non-degenerate solution has $m + n - 1$ allocations. S139–S141
- **NWCR**: start at the top-left cell, allocate min(supply, demand), strike the satisfied row or column, move on. Simple but may not be optimal. S142–S143

COST (RS)	KANPUR	PUNE	DELHI	SUPPLY
Jaipur	4	5	1	40
Udaipur	3	4	3	60
Mumbai	6	2	8	70
Demand	70	40	60	170

NWCR allocations: Jaipur–Kanpur 40, Udaipur–Kanpur 30, Udaipur–Pune 30, Mumbai–Pune 10, Mumbai–Delhi 60. Cost = 4×40 + 3×30 + 4×30 + 2×10 + 8×60 = **Rs 870**. S143–S149

LECTURE 50 S152–S193

Transportation Problem: Least Cost, VAM, MODI

- **Least Cost Method**: allocate to the cheapest cell first. Same data: Jaipur–Delhi (cost 1) gets 40 first. Cost = 1×40 + 3×40 + 3×20 + 6×30 + 2×40 = **Rs 480**. A tie at cost 3 gives an alternate solution of **Rs 520**. S154–S162
- **Vogel's Approximation Method (VAM)**, the penalty method: for each row and column, penalty = difference between the two lowest costs; pick the largest penalty; allocate to its cheapest cell; repeat. Here also **Rs 480**; VAM is generally the **most effective** of the three. S163–S167

MODI (Modified Distribution) optimality test

- Needs **m + n – 1** allocations in independent cells. S172
- For occupied cells set $\mathbf{u_i + v_j = c_{ij}}$ (take one variable = 0). For empty cells compute $\mathbf{d_{ij} = c_{ij} - (u_i + v_j)}$; u + v is the "implicit cost". S174–S177
- All d_{ij} positive → optimal; a zero → optimal with alternate solutions; any negative → improve via a closed loop (+θ/−θ). S178
- In the LCM solution: $u_3 = 0 \rightarrow v_2 = 2$, $v_1 = 6$, $u_1 = -5$, $v_3 = 6$, $u_2 = -3$; all vacant cells positive → **Rs 480 is optimal**. S175–S178

Unbalanced and second example

- Supply 117 > demand 95 → add a **dummy demand column of 22** with zero costs. If demand exceeded supply, add a dummy supply row. S179–S182
- Example 2 (P, Q, R × A–E; Σ = 340): LCM gives **Rs 1,410**; MODI finds $d_{11} = -1$, so one loop of θ = 10 improves it to the optimum **Rs 1,400**, with an alternate optimum ($d_{32} = 0$). S183–S189
- Advantages: competitive advantage, cost minimization, service levels, inventory management, resource optimization, time efficiency. S190–S191

LECTURE 51 S194–S221

Assignment Problem Programming

- A special case of the transportation problem: assign **n resources to n activities** one-to-one to minimize cost or maximize profit (workers to machines, salesmen to areas). S196–S197
- LP form: $x_{ij} = 1$ if resource i does task j, else 0; each task gets exactly one resource and each resource one task. S198–S199
- **Balanced**: square matrix. **Unbalanced**: rows ≠ columns → add a dummy row or column with **zero cost**. S200–S201

Hungarian method

1. Balance the cost table. S201
2. Row reduction (subtract each row's minimum), then column reduction. S201
3. Assign single zeros by rows then columns; strike other zeros in the same row/column. S201–S202
4. Optimal if assignments = number of rows. S202
5. Otherwise cover all zeros with the minimum number of lines (tick method). S202–S203
6. Subtract the smallest uncovered value K from uncovered cells, add K at line intersections. S203
7. Repeat until optimal. S203

EXAMPLE	RESULT
Balanced: 3 programmers × programmes A, B, C (minutes) S205–S207	1 → C (80), 2 → B (90), 3 → A (100); total **280 minutes**
Unbalanced: 4 jobs × 5 machines (dummy job 5) S208–S212	1 → B (3), 2 → A (10), 3 → D (1), 4 → C (6); machine **E gets no job**; total **Rs 20**

- Advantages: efficiency, clarity of objective, ease (binary variables, tables), flexibility, real-time applications (IoT, workforce, routing). S213–S217
- Assumptions: deterministic demand and supply, homogeneous tasks and resources, fixed costs, no transshipment, linear relationships, simple capacity limits, single period, non-negative variables, fixed origins and destinations. S218–S219

BEFORE THE EXAM ALL LECTURES

Cheat sheet

EMA	$\alpha = 2/(N+1)$; $EMA_t = \alpha Y_t + (1-\alpha)EMA_{t-1}$ S32
Regression	$b_1 = \Sigma(x-\bar{x})(y-\bar{y}) / \Sigma(x-\bar{x})^2$; $b_0 = \bar{y} - b_1\bar{x}$ S42
Efficiency · utilization	Actual ÷ effective · actual ÷ design S51
Break-even	FC ÷ (r − v); make-or-buy FC ÷ $(V_B - V_M)$ S61 S63
Sequencing	SPT minimizes flow time; LS = d − p; CR = time to due ÷ processing left (lowest first) S75–S80
Johnson	Shortest on M1 → first; on M2 → last; 3 machines → G = A + B, H = B + C S87 S91
Transport components	Flow, mode, load, route S100
Transportation LP	m + n − 1 allocations; NWCR → LCM → VAM (best); MODI: u + v = c, d = c − (u + v) S140–S177
Assignment	Hungarian; dummy rows/columns cost 0 S201

ANSWER TO REMEMBER	EXAMPLE
`y = 5x + 10`	TV ads vs cars S42
`38,640 · 64.7% · 89.3%`	Capacity example S52
`800 · 2,800 · 3,200`	Pizza BEP · for $10k profit · make-or-buy BEP S62–S65
`C-A-D-E-B`	Johnson 2-machine S90
`1-4-5-3-2 · 49`	Johnson 3-machine S94
`870 · 480 · 480`	NWCR · LCM · VAM (Jaipur example) S149–S167
`1,410 → 1,400`	LCM → MODI optimum S184–S189
`280 · Rs 20`	Assignment balanced · unbalanced S207 S212

MEMORY HOOK (MY OWN, NOT FROM THE SLIDES)

Johnson: "**F**irst machine → **F**ront, second machine → back". Initial solutions get smarter in alphabetical order of effort: **N**WCR (blind) < **L**CM (greedy) < **V**AM (penalties).

READ ONCE SOURCE CHECK

Slide quirks to watch

- **Pizza profit:** the slide gives profit at 1,200 pizzas as 1,200 × 9 − 1,200 × 4 = $6,000, leaving out the $4,000 fixed cost. With it, profit is **$2,000**. S62
- The WMA example uses only four months (Jan–Apr), unlike the six-month SMA and EMA examples. S37
- The Salesforce composite forecast is described as a feature of the Salesforce CRM platform rather than the classic "sales force composite" (sales reps' estimates). S20
- The critical ratio text says "Like DLS below", but no DLS rule follows. S80
- "Full shipment" is listed as a transportation-problem assumption ("either the entire quantity or none"), which the worked examples do not follow (they split supplies). S133
- "Vogle's Approximation Method" is Vogel's. S163
- Several slides are image-only or blank in the PDF: S6, S43–S45, S66–S68, S95–S97, S108–S109, S170–S171, S173, S220.

SELF-TEST 59 QUESTIONS

Week 10 practice questions

Answer these first, then check the key on the next pages. Option letters match the online quiz.

Q1 LECTURE 45

According to the World Bank's IDU (October 2023), India's growth in FY22/23 was:

- **A** 7.2%
- **B** 15.8%
- **C** 8.4%
- **D** 6.3%

Q2 LECTURE 45

In Q3 FY2024, GDP grew 8.4% while GVA grew 6.5%. The slides interpret this gap as:

- **A** a rise in agriculture
- **B** falling exports
- **C** demand growing faster than supply, risking inflation
- **D** supply outpacing demand

Q3 LECTURE 45

Passive demand forecasting is suitable for companies aiming for:

- **A** steady sales
- **B** rapid growth
- **C** new product launches
- **D** market entry

Q4 LECTURE 45

Long-term demand forecasting covers a period of:

- **A** less than a week
- **B** one month
- **C** more than a year
- **D** exactly one quarter

Q5 LECTURE 45

In the Delphi method, experts answer questionnaires independently mainly to:

- **A** reduce cost
- **B** avoid groupthink and the influence of dominant individuals
- **C** skip feedback rounds
- **D** save time

Q6 LECTURE 45

Which is a QUANTITATIVE demand forecasting technique?

- **A** Delphi method
- **B** Trend projection
- **C** Surveys
- **D** Focus groups

Q7 LECTURE 45

For a 3-period EMA, the smoothing factor $\alpha = 2/(N + 1)$ equals:

- **A** 0.67
- **B** 0.5
- **C** 0.33
- **D** 0.25

Q8 LECTURE 45

With $\alpha = 0.5$, EMA(Feb) = 155 and March sales = 170, EMA(March) is:

- **A** 170
- **B** 162.5
- **C** 165
- **D** 160

Q9 LECTURE 45

The 3-month SMA for June, using April 180, May 190 and June 200, is:

- **A** 180
- **B** 185
- **C** 200
- **D** 190

Q10 LECTURE 45

Using weights 1, 2, 3, 4 for Jan–Apr sales of 150, 160, 170, 180, the weighted moving average is:

- **A** 170
- **B** 172.5
- **C** 165
- **D** 175

Q11 LECTURE 45

For TV ads (1, 3, 2, 1, 3) and cars sold (14, 24, 18, 17, 27), the regression line is:

- **A** $y = 2x + 20$
- **B** $y = 4x + 12$
- **C** $y = 10x + 5$
- **D** $y = 5x + 10$

Q12 LECTURE 46

Capacity planning that bridges strategic and operational planning over a few months to a year is:

- **A** long-term
- **B** project
- **C** medium-term (tactical)
- **D** short-term

Q13 LECTURE 46

Effective capacity equals:

- **A** design capacity + allowances
- **B** design capacity – allowances
- **C** actual output × 2
- **D** utilization × efficiency

Q14 LECTURE 46

Efficiency is calculated as:

- **A** actual output ÷ effective capacity
- **B** actual output ÷ design capacity
- **C** effective ÷ design capacity
- **D** design ÷ actual

Q15 LECTURE 46

A plant makes 230 units/hour, runs 7 days with three 8-hour shifts. Weekly design capacity is:

- **A** 25,000
- **B** 38,640
- **C** 28,000
- **D** 5,520

Q16 LECTURE 46

With design capacity 38,640, effective capacity 28,000 and actual output 25,000, utilization is about:

- **A** 64.7%
- **B** 89.3%
- **C** 72.5%
- **D** 100%

Q17 LECTURE 46

The break-even quantity is:

- **A** $FC \times (r - v)$
- **B** $FC \div (r + v)$
- **C** $FC \div (r - v)$
- **D** $(r - v) \div FC$

Q18 LECTURE 46

Pizza: fixed cost $4,000/month, variable cost $4, price $9. Break-even quantity is:

- **A** 800
- **B** 1,000
- **C** 500
- **D** 444

Q19 LECTURE 46

For the same pizza line, how many pizzas give a $10,000 monthly profit?

- **A** 1,800
- **B** 2,000
- **C** 2,800
- **D** 3,200

Q20 LECTURE 46

ABX: make has FC $160,000 and $100/unit; buy costs $150/unit. The indifference quantity is:

- **A** 1,000
- **B** 1,600
- **C** 3,200
- **D** 5,000

Q21 LECTURE 46

At ABX's annual demand of 1,000 units, the company should:

- **A** buy ($150,000)
- **B** close the line
- **C** be indifferent
- **D** make ($260,000)

Q22 LECTURE 46

Which is a NON-economic consideration when evaluating capacity alternatives?

A Revenue
B Useful life
C Break-even analysis
D Public opinion

Q23 LECTURE 47

Scheduling differs from sequencing because it also determines:

A start and finish times
B inventory levels
C product prices
D only job order

Q24 LECTURE 47

In the A–E example, which rule gives the lowest total flow time (36 days)?

A SPT
B EDD
C FIFO
D LPT

Q25 LECTURE 47

Under FIFO in the example, the average delay per job is:

A 4.0 days
B 4.6 days
C 6.8 days
D 2.4 days

Q26 LECTURE 47

Which sequencing rule aims to reduce changeovers and increase machine utilization?

A EDD
B LPT
C FIFO
D SPT

Q27 LECTURE 47

Slack time for a job is calculated as:

A p – d
B d ÷ p
C p ÷ d
D d – p

Q28 LECTURE 47

Jobs with (time remaining, processing time): A (10, 5), B (15, 3), C (5, 2), D (7, 7). By critical ratio, which job goes first?

A B
B A
C C
D D

Q29 LECTURE 47

Johnson's rule gives an optimal polynomial-time solution for:

A assignment problems
B 1 job on n machines
C n jobs on 2 machines
D n jobs on any number of machines

Q30 LECTURE 47

In Johnson's rule, if the shortest time is on the SECOND work centre, that job is scheduled:

A in the middle
B first
C removed
D last

Q31 LECTURE 47

Jobs (M1, M2): A (3.2, 4.2), B (4.7, 1.5), C (2.2, 5.0), D (5.8, 4.0), E (3.1, 2.8). Johnson's optimal sequence is:

A A B C D E
B B E D A C
C C E A D B
D C A D E B

Q32 LECTURE 47

To use Johnson's rule for three machines A, B, C, the fictitious machines are:

A G = A + B, H = B + C
B G = A + C, H = B + C
C G = A – B, H = C – B
D G = A × B, H = B × C

Q33 LECTURE 47

In the MDH Masala three-machine example, the total elapsed time is:

A 40 hours
B 49 hours
C 35 hours
D 60 hours

Q34 LECTURE 48

The four major components of transportation decisions are:

A truck, rail, air, sea
B flow, mode, load, route
C cost, time, quality, safety
D plan, source, make, deliver

Q35 LECTURE 48

Distributing weight evenly in a vehicle using axle weight and centre of gravity is:

A palletization
B load composition
C load volume
D load balancing

Q36 LECTURE 48

Which mode has the lowest transportation cost per ton for long-haul bulk shipments?

A Road
B Air
C Pipeline only for solids
D Maritime

Q37 LECTURE 48

Which mode is best for time-sensitive, high-value and perishable goods but costs the most?

A Rail
B Air
C Pipeline
D Maritime

Q38 LECTURE 48

In the plant-to-warehouse mode choice example, which mode has the lowest total cost?

A Rail ($1,500,088)
B Air ($1,706,770)
C All equal
D Truck ($861,748)

Q39 LECTURE 48

US companies spend more than _____ on transportation every year.

A $600 billion
B $6 trillion
C $1 billion
D $60 billion

Q40 LECTURE 48

Tree of Life improved inbound freight efficiency from 55% to 75% using a TMS from:

A SAP
B HighJump
C LeanLogistics
D JDA Software

Q41 LECTURE 49

A transportation problem is balanced when:

A costs are equal
B m = n
C total supply equals total demand
D all allocations are equal

Q42 LECTURE 49

A basic feasible solution with m sources and n destinations should have how many allocations?

A m × n
B m + n
C m – n + 1
D m + n – 1

Q43 LECTURE 49

The North-West Corner Rule starts allocating from:

A the cell with the highest penalty
B the cheapest cell
C the top-left cell
D the bottom-right cell

Q44 LECTURE 49

Which cost ASSUMPTION underlies the transportation problem?

A Costs fall with volume
B Costs are zero
C Costs are random
D Cost per unit is constant (linear) regardless of quantity

Q45 LECTURE 49

For the Jaipur/Udaipur/Mumbai example, the NWCR transportation cost is:

A Rs 1,410
B Rs 520
C Rs 480
D Rs 870

Q46 LECTURE 50

The Least Cost Method allocates first to:

A the north-west cell
B the cell with the lowest transportation cost
C the dummy column
D the row with the largest penalty

Q47 LECTURE 50

For the same example, the Least Cost Method cost is:

A Rs 400
B Rs 480
C Rs 870
D Rs 520

Q48 LECTURE 50

In Vogel's Approximation Method, the penalty for a row is:

A its highest cost
B the difference between its two lowest costs
C the sum of its costs
D its supply

Q49 LECTURE 50

Which initial solution method is described as generally the most effective?

A Least Cost
B Vogel's Approximation Method
C North-West Corner
D Random allocation

Q50 LECTURE 50

In the MODI method, for each occupied cell:

A $u_i = v_j$
B $u_i + v_j = c_{ij}$
C $u_i \times v_j = c_{ij}$
D $u_i - v_j = c_{ij}$

Q51 LECTURE 50

In MODI, if a vacant cell's evaluation is negative, then:

A the problem is unbalanced
B there is an alternate optimum
C the solution is optimal
D the solution is not optimal and can be improved

Q52 LECTURE 50

Supply is 117 and demand 95. To balance, you add:

A nothing
B a dummy column of 117
C a dummy supply row of 22
D a dummy demand column of 22

Q53 LECTURE 50

In Example 2, the initial LCM cost of Rs 1,410 is improved by MODI to:

A Rs 1,410
B Rs 1,380
C Rs 1,400
D Rs 1,450

Q54 LECTURE 51

An assignment problem is a special case of the:

A transportation problem
B break-even problem
C sequencing problem
D queueing problem

Q55 LECTURE 51

An unbalanced assignment problem is made square by adding a dummy row or column with cost:

A zero
B infinity
C equal to the largest cost
D one

Q56 LECTURE 51

The first steps of the Hungarian method are:

A column then row reduction
B row reduction then column reduction
C draw lines then assign
D add K to all cells

Q57 LECTURE 51

In the Hungarian method, if the minimum number of lines covering all zeros is less than the matrix order, you:

A add a dummy row
B stop; it's optimal
C restart from scratch
D subtract the smallest uncovered value from uncovered cells and add it at intersections

Q58 LECTURE 51

In the balanced programmer example, the minimum total computer time is:

A 300 minutes
B 250 minutes
C 280 minutes
D 270 minutes

ANSWERS 59 QUESTIONS

Answer key

Q	ANS	WHY · SLIDE
Q1	A	Second highest among G20 countries; the FY23/24 forecast was 6.3%. S3
Q2	C	Excessive demand for too few goods. S5
Q3	A	Active forecasting suits startups and fast-growing firms. S15
Q4	C	S16. S16
Q5	B	Typically 2–4 rounds until consensus. S19
Q6	B	Quantitative: moving averages, trend projection, seasonal averages, econometrics, ML. S17
Q7	B	2/(3 + 1) = 0.5. S32
Q8	B	0.5 × 170 + 0.5 × 155 = 85 + 77.5 = 162.5. S33
Q9	D	(180 + 190 + 200) ÷ 3 = 190. S30
Q10	A	(150 + 320 + 510 + 720) ÷ 10 = 1700 ÷ 10 = 170. S37
Q11	D	$\bar{x} = 2$, $\bar{y} = 20$, $\Sigma(x-\bar{x})(y-\bar{y}) = 20$, $\Sigma(x-\bar{x})^2 = 4 \rightarrow b_1 = 5$, $b_0 = 10$. S42
Q12	C	S49. S49
Q13	B	S51. S51
Q14	A	Utilization uses design capacity. S51
Q15	B	7 × 3 × 8 × 230 = 38,640. S52
Q16	A	25,000 ÷ 38,640 = 64.7%; efficiency = 25,000 ÷ 28,000 = 89.3%. S52
Q17	C	S61. S61
Q18	A	4,000 ÷ (9 − 4) = 800. S62
Q19	C	(10,000 + 4,000) ÷ 5 = 2,800. S62
Q20	C	160,000 ÷ (150 − 100) = 3,200. S65
Q21	A	Make costs 100,000 + 160,000 = 260,000; buy costs 150,000. Make is better only above 3,200 units. S65
Q22	D	Also reactions from employees and community pressure. S59
Q23	A	S71. S71
Q24	A	FIFO 50, LIFO 46, SPT 36, LPT 60, EDD 39. S75
Q25	B	(0 + 1 + 2 + 6 + 14) ÷ 5 = 4.6. S73
Q26	B	S76. S76
Q27	D	s = due date − processing time. S79
Q28	D	CR: A 2.0, B 5.0, C 2.5, D 1.0 → D, A, C, B. S81–S82
Q29	C	General flow shop is exponential. S86
Q30	D	Shortest on the first work centre → first. S87
Q31	D	B last (1.5 on M2), C first (2.2 on M1), E last (2.8 on M2), A first (3.2 on M1), D remains. S90
Q32	A	Valid if min(A) ≥ max(B) or min(C) ≥ max(B). S91
Q33	B	Sequence 1, 4, 5, 3, 2. S94
Q34	B	S100. S100
Q35	D	S104. S104
Q36	D	S113. S113
Q37	B	S114. S114
Q38	D	Total includes transport, in-transit, plant and warehouse inventory costs. S118
Q39	A	About $1 billion more is spent on TMS. S125
Q40	D	Polaroid used HighJump; Orange Glo used LeanLogistics. S126
Q41	C	Otherwise add a dummy source or destination. S138
Q42	D	S140 and S172. S140
Q43	C	S142. S142
Q44	D	Cost linearity. S132
Q45	D	4×40 + 3×30 + 4×30 + 2×10 + 8×60 = 870. S149
Q46	B	S154. S154
Q47	B	1×40 + 3×40 + 3×20 + 6×30 + 2×40 = 480. The tie alternative gives 520. S160
Q48	B	Pick the row/column with the largest penalty. S163
Q49	B	S167. S167

Q	ANS	WHY · SLIDE
Q50	**B**	For empty cells $d_{ij} = c_{ij} - (u_i + v_j)$. S174
Q51	**D**	Zero → optimal with alternate solutions; all positive → optimal. S178
Q52	**D**	Supply > demand → dummy demand column with zero costs. S181
Q53	**C**	$d_{11} = -1$ → one loop with $\theta = 10$. S189
Q54	**A**	n resources to n activities, one-to-one. S196
Q55	**A**	S201. S201
Q56	**B**	Subtract each row's minimum, then each column's minimum. S201
Q57	**D**	Cells covered by one line stay unchanged. S203
Q58	**C**	1 → C (80), 2 → B (90), 3 → A (100). S207
Q59	**A**	1 → B (3), 2 → A (10), 3 → D (1), 4 → C (6) = 20. S212

Made from *WEEK 10.pdf* (221 slides, NPTEL Logistics & Supply Chain Management, Prof. Vikas Thakur, IIT Kharagpur). Notes use only the slides, except boxes marked "my own".